Tarea

2.23) a) CH_4 = $C = 12.011$
$H = 1.00794 * 4 = 4.03$
$CH_4 = 16.04$ uma

b) NO_2 = $N = 14.0067$
$O = 15.9994 * 2 = 32$ = 46.01 uma

c) SO_3 = $S = 32.066$
$O = 15.9994 * 3 = 48$ = 80.07 uma

d) C_6H_6 = $C = 12.011 * 6 = 72.07$ = 78.12 uma
$H = 1.00794 * 6 = 6.05$

e) NaI = $Na = 22.98977$ = 149.90 uma
$I = 126.9045$

f) K_2SO_4 = $K = 39.0983 * 2 = 166.0028$
$S = 32.066$ = 262.07 uma
$O = 15.9994 * 4 = 64$

g) $Ca_3(PO_4)_2$ = $Ca = 40.078 * 3 = 120.234$
$P = 30.97376$ = 215.21 uma
$O = 15.9994 * 4 = 64$

2.24) a) Li_2CO_3 = Li = 6.941 * 2 = 13.90
C = 12.011
O = 15.9994 * 3 = 48
= 74 g/mol

b) CS_2 = C = 12.011
S = 32.066 * 2 = 64.132
= 76.14 g/mol

c) $CHCl_3$ = C = 12.011
H = 1.00794
Cl = 35.4527 * 3 = 106.3581
= 119.41 g/mol

d) $C_6H_8O_6$ = C = 12.011 * 6 = 72.066
H = 1.00794 * 8 = 8.06352
O = 15.9994 * 6 = 95.9964
= 176.13 g/mol

e) KNO_3 = K = 39.0983
N = 14.0067
O = 15.9994 * 3 = 47.9982
= 101.10 g/mol

f) Mg_3N_2 = Mg = 24.305 * 3 = 72.915
N = 14.0067 * 2 = 28.134
= 101.05 g/mol

3.25) $152 g/mol * \frac{7g}{0.372mol} = 408.60g$

$152 g * \frac{1mol}{408.60g} = \underline{0.37mol}$

3.26) $C_2H_6 = 30.07g$

$0.334gC_2H_6 * \frac{1mol}{30.07ggC_2H_6} * \frac{1mol}{}$

3.27) $C_6H_{12}O_6 = 180.16g$

$1.50gC_6H_{12}O_6 * \frac{1mol}{180.16g} * \frac{6mol C}{1mol C_6H_{12}O_6} * \frac{6.022 \times 10^{23} atom C}{1mol C} = 3 \times 10^{22} atom C$

$1.50gC_6H_{12}O_6 * \frac{1mol}{180.16g} * \frac{12mol H}{1mol C_6H_{12}O_6} * \frac{6.022 \times 10^{23} atom H}{1mol H} = 6.02 \times 10^{22} atom H$

$1.50gC_6H_{12}O_6 * \frac{1mol}{180.16g} * \frac{6mol O}{1mol C_6H_{12}O_6} * \frac{6.022 \times 10^{23} atom O}{1mol O} = 3 \times 10^{22} atom O$

3.28) $[(CH_3)_2SO] = 63.34g$

$7.14 \times 10^3 g * \frac{1mol}{63.34g} * \frac{2mol C}{1mol (CH_3)_2SO} * \frac{6.022 \times 10^{23} atom C}{1mol C} = 1.36 \times 10^{26} atom C$

$$7.14\times10^{3}g * \frac{1mol}{63.37g} * \frac{1mol S}{1mol[(CH_2)_2S]} * \frac{6.022\times10^{23}\,atom S}{1mol S} = 6.79\times10^{25}\,atom S$$

$$7.14\times10^{3}g * \frac{1mol}{63.37g} * \frac{3mol H}{1mol[(C_2H_2)_2SO]} * \frac{6.022\times10^{23}\,atom H}{1mol H} = 2.08\times10^{26}\,atom H$$

$$7.14\times10^{3}g * \frac{1mol}{63.37g} * \frac{1mol O}{1mol[(C_2H_2)_2SO]} * \frac{6.022\times10^{23}\,atom O}{1mol O} = 6.79\times10^{25}\,atom O$$

3.29) $C_{19}H_{38}O$

3.30) $H_2O = 18g$

$$2.56ml\,H_2O * \frac{1g\,H_2O}{1ml\,H_2O} * \frac{1mol\,H_2O}{18g\,H_2O} * \frac{6.022\times10^{23}\,mol\,H_2O}{1mol\,H_2O} = 8.56\times10^{22}\,mol\,H_2O$$

3.31) Mide razones masa/carga de iones, calentando un haz de material del compuesto a analizar hasta vaporizarlo e ionizar los diferentes átomos.

3.32) Se puede medir o determinar mediante una técnica llamada espectrometría de masas.

3.33)

CF_4

$\frac{126\,C}{12\,C}$ $\frac{15\,C}{6\,C}$ = 98.9%

3.34)

$\frac{34\,S}{16\,S}$ $\frac{36\,S}{6\,S}$ = 104.3%

3.34)

$$\frac{34\,g}{16\,g} \quad \frac{36\,g}{6\,g} = 104.3\%$$

3.35) La composición porcentual es una medida de la cantidad de masa que ocupa un elemento en un compuesto. Se mide en porcentaje de masa.

3.36) Para poder identificarlo es un poco más fácil con la fórmula $= \frac{\text{peso atómico} \times \text{n átomos en la molécula}}{\text{peso molecular}} \times 100$

3.37) Es una expresión que representa los átomos que forman un compuesto químico sin atender a su estructura.

3.38) Se necesita el peso molecular y el número de átomos.

3.39) SnO_2 =

Sn = 118.710 = 118.710 ÷ 150.7088 = 0.80 * 100 = 80% Sn
O = 15.9994 * 2 = 31.9988 ÷ 150.7088 = 0.20 * 100 = 20% O
150.7088 g/mol

3.40) $CHCl_3$

C = 12.011 = 12.011 ÷ 119.37704 = 0.10 * 100 = 10% C
H = 1.00794 = 1.00794 ÷ 119.37704 = 8.44×10^{-3} * 100 = 0.84% H
Cl = 35.4527 * 3 = 106.3581 ÷ 119.37704 = 0.89 * 100 = 89% Cl
119.37704

3.41) $C_9H_{10}O$

C = 12.011 * 9 = 108.099 ÷ 134.1778 = 0.81 * 100 = 80% C
H = 1.00794 * 10 = 10.0794 ÷ 134.1778 = 0.08 * 100 = 8% H
O = + 15.9994 ÷ 134.1778 = 0.12 * 100 = 12% O
134.1778

3.42)

a) $(NH_2)_2CO$

N = 14.0067 ÷ 44.03298 = 0.32 * 100 = 32% N
H = 1.00794 * 2 = 2.01588 ÷ 44.03298 = 0.05 * 100 = 5%
C = 12.011 ÷ 44.03298 = 0.27 * 100 = 27%
O = + 15.9994 ÷ 44.03298 = 0.36 * 100 = 36%
44.03298

b) NH_4NO_3

N = 14.0067 * 2 = 28.01 ÷ 80.01614 = 0.35 * 100 = 35% N
H = 1.00794 * 4 = 4.00794 ÷ 80.11614 = 0.05 * 100 = 5%
O = 15.9994 * 3 = 47.9982 ÷ 80.7614 = 0.60 * 100 = 60%
80.01614

c) $HNC(NH_2)_2$

H = 1.00794 * 3 = 3.00794 ÷ 43.03 = 0.07 * 100 = 7%
N = 14.0067 * 2 = 28.01 ÷ 43.03 = 0.65 * 100 = 65% N
C = + 12.011 ÷ 43.03 = 0.28 * 100 = 28%
43.03

d) NH_3 → Es la que tiene más porcentaje de nitrogeno

N = 14.0067 ÷ 17.01 = 0.82 * 100 = 82% N
H = 1.00794 * 3 = 3.00794 ÷ 17.01 = 0.18 * 100 = 18%
17.01

3.43)

C 44.4% = 44.4g $\times \frac{1\,mol}{12.011g}$ = 3.70 6

H 6.21% = 6.21g $\times \frac{1\,mol}{1.00794g}$ = 6.16 10

S 39.5% = 39.5g $\times \frac{1\,mol}{32.066g}$ = 1.23 2

O 9.86% = 9.86g $\times \frac{1\,mol}{15.9994g}$ = 0.62 1

FE = $C_6H_{10}S_2O$

3.44)

C = 19.8% = 19.8g $\times \frac{1\,mol}{12.011g}$ = 1.65 2

H = 2.50% = 2.50g $\times \frac{1\,mol}{1.00794g}$ = 2.48 3

N = 11.6% = 11.6g $\times \frac{1\,mol}{14.0067g}$ = 0.83 1

FE = C_2H_3N

Escaneado con CamScanner

3.45) $Fe_2O_3 = 159.70\,g$

$$24.6\,g \ \cancel{\times}\ \frac{1\,mol}{159.70\,g} = 0.15\,mol\,Fe$$